Total no. of Pages: .... Roll no. ....

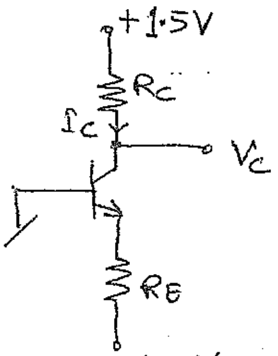
Degree: **B.Tech.(ECE-AIML)**, Semester: **IIIrd**
MID-SEMESTER EXAMINATION, September, 2024

Course Title: **Microelectronics Circuits and Applications**
Course Code: **EAEPC304**

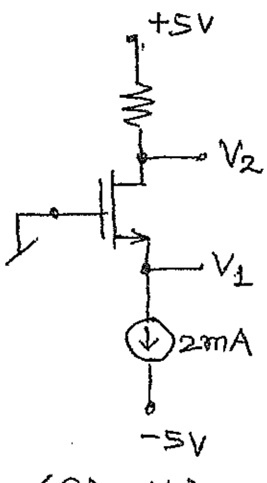
Duration: **1:30 Hours** Max. Marks: **20**

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Draw the neat output characteristic of an *npn* BJT, explain and exhibit the effect of base-width modulation thereon.	2	CO1
1b	An *npn* transistor's J_{CB} is reverse biased and emitter is left open circuited. Utilising Ebbers-Moll model description of the BJT determine the mode of its operation and the values of I_C, I_B and V_{EB}.	2	CO1
2a	Derive the expression for the transconductance of the MOSFET in saturation region and draw the low-frequency small-signal model.	2	CO1
2b	Derive the A_v, A_i, R_i, R_o of the CE and CB configurations and compare their typical magnitudes.	2	CO1
3a	With neat circuit diagrams and necessary derivations, discuss BJT and MOSFET based simple current mirrors. Compare their features too.	2	CO2
3b	Discuss the miller effect and derive the f_H of the common-emitter BJT configuration. (Use the hybrid-Pi model for high frequency)	2	CO2
4a	Design the circuit given in Fig.4a to establish the I_{CQ}=0.2mA and V_C=0.5V. The transistor exhibits v_{BE} of 0.8V at i_C=1mA and β=100.	2	CO2
4b	For the circuit given in Fig.4b calculate the DC node voltages V_1 and V_2 for V_t=1V and $k'(W/L)$=4mA/V^2 and λ=0.	2	CO3
5a	Calculate Av, Ri and Ro for the circuit given in Fig.5a	2	CO3
5b	Derive the transconductance (g_m) of the BJT differential pair and plot the v_{din} versus $I_{diffout}$ characteristic of the same.	2	CO3



(Fig.4a)



(Fig.4b)

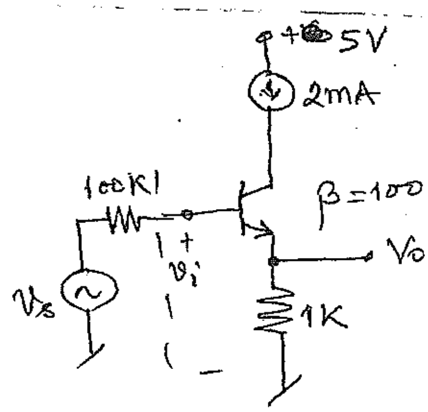


Fig.5a